

IILM UNIVERSITY

Department of Computer Science and Engineering

INTERNSHIP REPORT

ON

AI-POWERED BUSINESS AUTOMATION ASSISTANT USING LLM INTEGRATION AND MACHINE LEARNING

Submitted in partial fulfilment of the requirements for the degree of
Bachelor of Technology in Computer Science and Engineering

Submitted By

Name of Student: ROUNAK GUPTA

Roll Number: 25SCS1003005151

Batch: 2024–2028

Internship Organization: ADVLST (IT Solutions & AI Automation Company)

Internship Duration: 25 May 2026 – 10 July 2026

Academic Session 2026–27

CANDIDATE'S DECLARATION

I, ROUNAK GUPTA, do hereby declare that the internship report titled “AI-Powered Business Automation Assistant Using LLM Integration and Machine Learning” has been prepared by me to fulfil the academic requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering. The report is based on the internship learning and project work carried out by me during my tenure at ADVLST. References used in preparing the report have been acknowledged in the bibliography.

I further declare that this report has not been submitted to any other institution for the award of any degree or diploma. I shall be responsible for any copyright or academic-integrity violation arising from material submitted under my name.

Student Name – ROUNAK GUPTA

Roll No. – 25SCS1003005151

ACKNOWLEDGEMENT

I sincerely express my gratitude to everyone who supported me during the internship and the preparation of this report. The internship provided an opportunity to understand practical concepts related to Artificial Intelligence, Machine Learning, Large Language Model (LLM) integration, prompt engineering and AI-driven business automation.

I am thankful to ADVLST for providing the internship learning environment and project-oriented exposure in the AI Research & Development Division. I am especially grateful to Mr. Deepak, HR Head, ADVLST, and Mr. Akshat, Chief Director, ADVLST, for their guidance, support and encouragement throughout the internship period. I also acknowledge the guidance and academic support provided by my university and faculty members at IILM University. I am grateful to my parents and everyone who encouraged me throughout the internship period.

Student Name – ROUNAK GUPTA

Roll No. – 25SCS1003005151

INTERNSHIP COMPLETION CERTIFICATE

The Certificate of Internship issued by ADVLST is reproduced below. The complete offer letter and certificate are also attached in the “Certificate and Communication Proof” section of this report.



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4. PROJECT DESCRIPTION

4.1 Introduction

Artificial Intelligence and Machine Learning are increasingly used by organizations to automate repetitive tasks, analyse data and support decision-making. A major recent development in this area is the use of Large Language Models (LLMs), which can understand and generate natural language, and can be integrated into business applications through APIs to build intelligent assistants and automation systems.

During the internship at ADVLST, the project focused on developing an AI-powered automation assistant that combines traditional machine-learning techniques with LLM integration. The system was designed to process data, respond to natural-language requests, and support enterprise workflow automation using prompt engineering and API-based integration with LLM services.

The project follows a practical AI-development workflow: requirement understanding, data collection and preprocessing, model/prompt design, integration with LLM APIs, testing, and evaluation of outputs.

4.2 Organization Profile

ADVLST is an IT Solutions & AI Automation Company based in New Delhi, India, working in the areas of software infrastructure, backend systems and automation solutions. The AI & Machine Learning internship was carried out under the AI Research & Development Division of the organization, which focuses on building machine-learning models, LLM-based integrations and intelligent business-automation systems.

Particular	Details
Organization	ADVLST
Role	AI & Machine Learning Intern
Department	AI Research & Development Division
Duration	25 May 2026 – 10 July 2026
Mode of Internship	Remote / Work From Home
Working Schedule	Monday to Friday
Reference ID	ADV-INT-2026-152

Table 4.1: Internship Details

Internship Training Areas

Area	Key Learning
Python	Programming and implementation for AI applications
Machine Learning	Developing, testing and optimizing ML models
Large Language Models (LLMs)	Integration of LLMs via APIs into applications
Prompt Engineering	Designing effective prompts for reliable LLM outputs
Data Analysis	Analysing datasets and constructing data pipelines

API Integration	Connecting AI models/services with backend systems
Business Automation	Designing AI-driven automation for workflows

Table 4.2: Internship Training Areas

4.3 Problem Statement

Modern enterprises handle large volumes of repetitive queries, data-processing tasks and communication workflows that consume significant manual effort. Rule-based automation is often rigid and cannot handle the variability of natural-language requests or unstructured data.

The problem addressed by this project was to design and build an AI-driven assistant capable of interpreting natural-language requests, applying machine-learning models where structured prediction was required, and using LLM integration to generate accurate, context-aware responses that support business automation workflows.

4.4 Project Objectives

- Understand the role of Machine Learning and LLMs in business automation.
- Design and preprocess datasets required for model development.
- Build and evaluate machine-learning models for relevant prediction/classification tasks.
- Integrate Large Language Models via APIs for natural-language understanding and generation.
- Apply prompt-engineering techniques to improve the reliability of LLM outputs.
- Design data pipelines connecting data sources, models and the automation layer.
- Evaluate the system's outputs using appropriate performance measures.
- Document the complete workflow and identify areas for future enhancement.

4.5 Scope of the Project

The project focuses on a manageable prototype of an AI-powered automation assistant, combining classical machine-learning components with LLM-based natural-language processing.

Included	Not Included
Data collection and preprocessing	Full enterprise-scale deployment
ML model development and evaluation	Production-grade infrastructure and scaling
LLM API integration and prompt engineering	Fine-tuning of proprietary large-scale LLMs
Prototype automation workflow	Guaranteed accuracy on all real-world edge cases

4.6 Technologies and Tools Used

Technology / Tool	Purpose
Python	Main programming language
Pandas / NumPy	Data handling and numerical operations
Scikit-learn	Machine-learning model implementation
LLM APIs (via Prompt Engineering)	Natural-language understanding and generation
REST APIs	Integration between models, data and application layer
Jupyter Notebook	Development and experimentation
Matplotlib	Data visualization

Table 4.3: Technologies and Tools Used

4.7 System Architecture

The proposed system consists of a data/ML pipeline and an LLM-based response-generation pipeline, connected through a common automation layer.

User Request → Preprocessing → [ML Model | Prompt Construction + LLM API] → Response Generation → Automation Action / Output

Figure 4.1: Proposed System Architecture

Structured requests (e.g., classification or prediction tasks) are routed to the trained machine-learning model, while natural-language requests are converted into an engineered prompt and sent to the LLM API. Outputs from either path are passed through a response-generation layer before being delivered to the end user or triggering an automation action.

4.8 Methodology

- Requirement Understanding
- Data Collection and Inspection
- Data Cleaning and Preprocessing
- Feature Engineering (for ML components)
- Model Training and Evaluation
- Prompt Design and LLM API Integration
- Testing of End-to-End Workflow
- Error Analysis and Refinement
- Documentation and Future Enhancement

Figure 4.2: Overall Project Methodology

Data Preprocessing

Data used for the ML components was inspected for missing values, duplicates and class imbalance, and cleaned before feature extraction. For LLM-based components, input text was normalized and structured into consistent prompt templates to ensure reliable outputs.

Figure 4.3: Data & Prompt Processing Workflow

Model Training and Evaluation

Baseline machine-learning models were trained and evaluated using standard classification metrics. LLM outputs were evaluated qualitatively for relevance, correctness and consistency, and iteratively refined through prompt adjustments.

Measure	Purpose
Accuracy	Overall proportion of correct predictions
Precision	Reliability of positive predictions
Recall	Ability to identify relevant positive cases
F1-score	Combined measure of precision and recall
Response Relevance (LLM)	Qualitative assessment of LLM output usefulness

Table 4.4: Model Evaluation Measures

New Request Handling Workflow

**New User Request → Preprocessing → Routing (ML / LLM) → Prompt Construction (if LLM) →
Model / API Call → Response Delivered**

Figure 4.4: New Request Handling Workflow

After development, a new request is passed through the same preprocessing and routing logic used during development, ensuring consistency between the design-time and run-time behaviour of the system.

4.9 Expected Outcomes / Learning Outcomes

- A working prototype combining ML models and LLM-based natural-language processing.
- A structured data pipeline for preprocessing and feature handling.
- Practical experience with prompt engineering and LLM API integration.
- Trained and evaluated machine-learning model(s) for the chosen task.
- Understanding of end-to-end AI-driven automation workflow design.
- Analysis of system outputs and identification of improvement areas.
- A foundation for future enhancement such as advanced fine-tuning or scaling the automation system.

The internship also strengthened understanding of the complete AI-application development lifecycle, including the importance of well-designed prompts, clean data, and consistent processing between development and deployment.

4.10 Certificate and Communication Proof

The internship offer letter and completion certificate issued by ADVLST for Rounak Gupta are reproduced below without altering their substantive content.



ADVLST

IT Solutions & AI Automation Company
Software Infrastructure • Backend Systems • Automation Solutions
New Delhi, India
www.advlst.in

Date: 8 May 2026
Reference ID: ADV-INT-2026-152
Internship Type: Remote
Department: AI Research & Development

INTERNSHIP OFFER LETTER

Dear **Rounak Gupta**,

We are pleased to offer you the position of **AI & Machine Learning Intern** at **ADVLST**.

During your internship tenure, you will work closely with our core engineering team on advanced artificial intelligence projects, including the development of machine learning models, LLM integrations, and intelligent automation systems tailored for modern enterprise workflows.

Intern Name	Rounak Gupta
Internship Role	AI & Machine Learning Intern
Department	AI Research & Development Division
Internship Duration	25 May 2026 – 10 July 2026
Working Schedule	Monday to Friday
Mode of Internship	Remote / Work From Home
Compensation	Performance-Based

Roles & Responsibilities

- Develop, test, and optimize machine learning models for internal applications.
- Assist in prompt engineering and integrating Large Language Models (LLMs) via APIs.
- Analyze complex datasets and construct robust data pipelines for AI training.
- Collaborate with the development team to design AI-driven business automation solutions.
- Conduct technical research on emerging AI frameworks and implementation strategies.

Technologies & Exposure

Machine Learning

Large Language Models (LLMs)

Python & AI Frameworks

Prompt Engineering

Data Analysis

API Integrations

This internship is highly specialized and is designed to provide you with practical, industry-level exposure to advanced artificial intelligence systems and state-of-the-art machine learning environments.

We look forward to your contribution and wish you a productive and valuable learning experience with our organization.


Deepak
HR Head, ADVLST

This is a digitally generated internship offer letter issued by ADVLST.
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Certificate of Internship

Ref: ADV-INT-2026-152



CERTIFICATE OF INTERNSHIP

THIS DOCUMENT IS PROUDLY PRESENTED TO

Rounak Gupta

In recognition of his successful completion of the **AI & Machine Learning** internship program at ADVLIST Strategy. During his tenure, Rounak demonstrated exceptional analytical skills, technical proficiency in developing machine learning models, and a strong commitment to delivering high-quality AI-driven solutions.

We deeply appreciate his dedication and wish him continued success in all future professional endeavors.

For verification mail us at
work@advlist.in


CHIEF DIRECTOR

5. CONCLUSION

The internship at ADVLST provided practical, industry-level exposure to Artificial Intelligence and Machine Learning, with a specific focus on integrating Large Language Models into business-automation workflows. The project reinforced classroom concepts of machine learning while introducing newer, industry-relevant practices such as prompt engineering and LLM API integration.

The internship helped build a strong foundation for future work in AI-driven application development and highlighted the importance of combining traditional ML techniques with modern generative-AI capabilities to build practical, real-world solutions.

6. BIBLIOGRAPHY / REFERENCES

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